Thermal-Induced Threshold Voltage Distribution Changes in 3D NAND Flash Cells: Experimental Characterization

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Abstract

Non-volatile memory devices are essential for persistent data storage, with NAND flash memory being a dominant technology. However, NAND flash memory suffers from reliability challenges, including threshold voltage (VT) drift due to temperature variations. In this work, we investigate the shift in VT distribution of NAND flash cells upon heating, characterizing the impact of annealing on cell performance. Experimental results demonstrate a clear shift in the VT distribution towards lower voltages after heating, which aligns with expectations and has significant implications for data retention and error rates.

Experimental Setup

To investigate the impact of heating on NAND flash memory cells, a controlled experimental setup was employed. Individual MT29F256G08EBHAF NAND flash memory chips were placed on a hot plate for annealing at a specific temperature (80°C) for 1 hour. The hot plate maintains a consistent temperature across the chip surface, ensuring uniform heating. An FTDI FT2232 H Mini module using the Bit Bang mode was used to program and read individual cells. The programming was performed on the pages of different blocks in the NAND flash memory. The chip was programmed using conventional methods, and the threshold voltages of cells within an SLC page were measured before and after heating. Post-heating cell characteristics were analyzed using the read operation on the pages of different blocks with sweeping of different voltage references to account for threshold voltage shifting. All data was recorded and analyzed on a computer using dedicated software. A flow chart in Figure 1(b) demonstrates this process.

Results and Discussion

The effect of heating on the threshold voltage (VT) distribution of NAND flash cells was thoroughly examined. The experimental results reveal a distinct shift in the VT distribution toward lower voltages upon annealing. Figure 2(a) depicts a simplified representation of electron leakage from the floating gate, emphasizing the increased electron loss at elevated temperatures and reduced margin for data distinction. The cumulative distribution functions (CDFs) of cell error counts versus threshold voltage shift, illustrating the effect of annealing (heating) on the performance of NAND flash memory are shown in figure 2(b). Figure 3(b) illustrates the pre- and post-annealing VT distributions, fitted with Gaussian curves. As depicted, the Gaussian distribution shifts leftward and widens post-annealing, which directly corresponds to the electron leakage mechanism illustrated in Figure 3(a). The leftward shift indicates a reduction in the average threshold voltage required to activate the cell, directly resulting from the loss of trapped electrons from the floating gate. Furthermore, the widening of the distribution is indicative of an increased variability in the threshold voltages of individual cells after thermal stress. This widening is attributed to non-uniformity in the leakage rates among different cells, potentially due to variations in oxide quality, trap densities, or local thermal conditions. A VT shift of approximately 31 mV was observed after heating. These observations align with findings in the research on non-volatile memories and suggest that thermal excitation accelerates charge loss mechanisms. The resulting threshold voltage shift compromises data retention and increases raw bit error rates, making temperature compensation techniques imperative in NAND flash memory systems.

References

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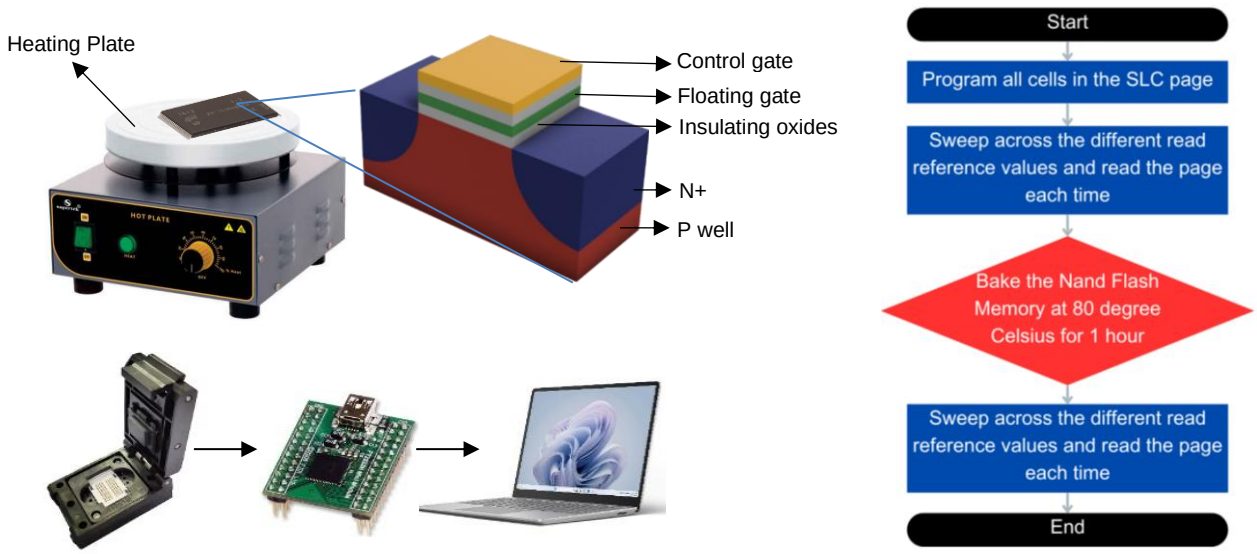

Fig. 1: (a) Experimental setup for characterizing the effects of heating on NAND flash memory. (b) Procedure for measuring the effect of baking the NAND flash memory cells at high temperatures.

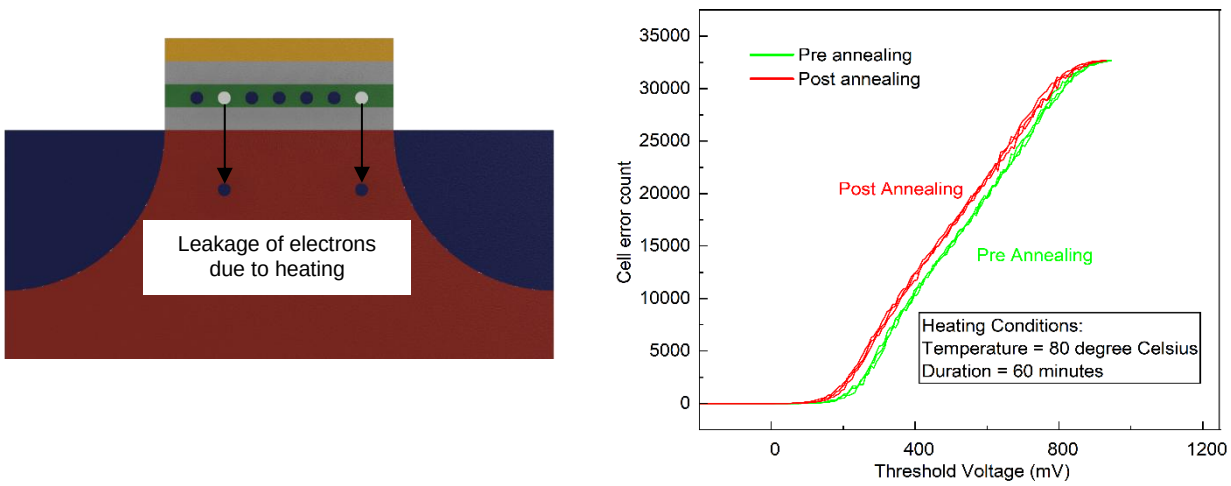

Fig. 2: (a) Illustration of electron leakage in a NAND flash cell due to heating. (b) Cumulative distribution functions (CDFs) of cell error counts for 6 different pages versus threshold voltage, illustrating the effect of annealing (heating) on the performance of NAND flash memory.

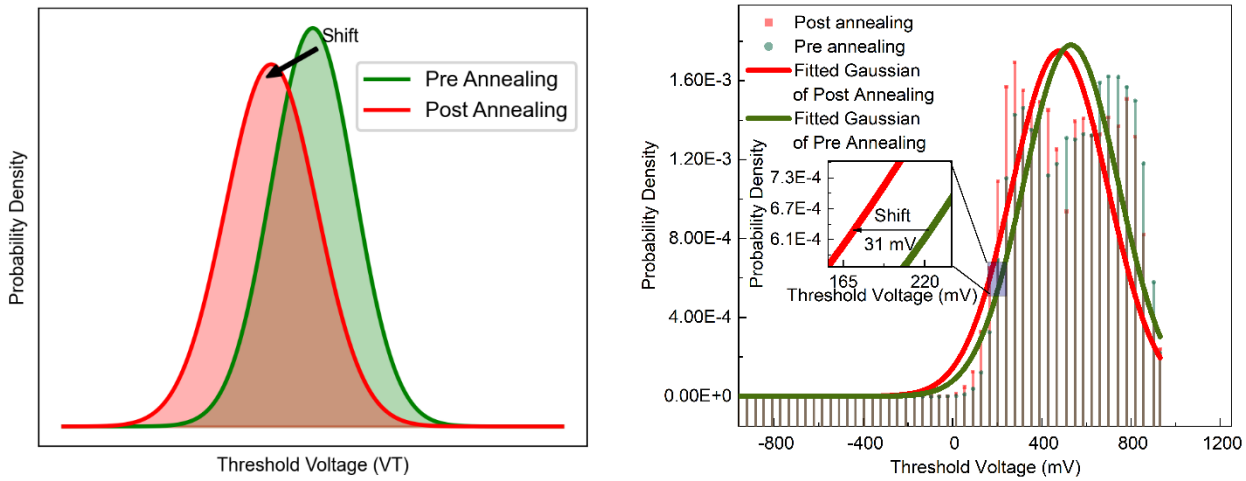

Fig. 3: (a) Graphical representation of the threshold voltage distribution in NAND flash memory cells, comparing the "Pre Annealing" (green) and "Post Annealing" (red) states. The shift in the distribution indicates a reduction in threshold voltage due to the annealing process. (b) Gaussian fit illustrating the shift in threshold voltage (VT) distribution of NAND flash cells after annealing, with an inset showing the VT shift of 31 mV.